

Proud Mqondisi Mathuthu

Johannesburg +27631737674 | mqondisimathuthu@gmail.com

<https://www.linkedin.com/in/mqondisi-mathuthu-5a5475158/> | <https://mqondisiproud.netlify.app/>

PROFESSIONAL SUMMARY

CCNP-level Network Engineer with **5 years of hands-on infrastructure experience** and a Bachelor of Science in Information Technology. Holds **CCNA** and **CCNP Service Provider** (Advanced Routing) - demonstrating expert-level command of BGP, OSPF, MPLS, VPNv4, and carrier-grade architectures. Strong automation practitioner (Python + Ansible) with production experience on **Fortinet** firewalls (FortiGate/FortiManager). Early adopter of AI-assisted workflows for troubleshooting, code generation, and documentation. On-site ready, travel-flexible, and accustomed to on-call and change-window rotations.

CERTIFICATIONS

- **Service Provider (CCNP SP)** - Advanced Routing / MPLS / VPNs – August 2024
- **Cisco Certified Network Associate (CCNA)** – October 2024
- **Fortinet NSE 4** - practical experience; certification in progress

PROFESSIONAL EXPERIENCE

Network Engineer | Brilliant Telecommunications | Randburg

[February, 2022 – Present]

- Manage a **hybrid SP/enterprise network** spanning 200+ devices (Cisco ASR/ISR/Nexus, FortiGate firewalls) across 5 regional PoPs.
- **Routing & SP expertise:** Designed and maintained eBGP peering with 3 upstream ISPs (full tables + default route), OSPF as internal IGP, and MPLS LDP for label-switched traffic. Deployed VPNv4 with route-targets for customer VRF segregation.
- **Fortinet security:** Administer FortiGate 600F/200F HA clusters - policies, SSL deep inspection, IPS/IDS profiles, and IPsec VPNs to branch sites. Migrated 20+ legacy Cisco ACLs to FortiGate object-based policies with application control.
- **Automation:** Built a Python library (netmiko) to push BGP prefix-lists and route-maps across 40+ routers in under 3 minutes (previously 2 hours manual). Wrote Ansible playbooks for version-controlled FortiGate policy backups using the fortios collection.
- **Load balancing:** Deployed HAProxy for internal web portals with SSL termination, custom health checks, and stick-tables; integrated with Prometheus for real-time metrics.
- **AI workflow:** Use GitHub Copilot to accelerate Python development; leverage AI assistants to translate BGP outputs into plain-English summaries during outage calls - reduced bridge time by 25%.
- Active participant in weekly change windows (22:00–02:00) and monthly on-call rotation; lead post-incident reviews and root-cause analysis.

Network Engineer Intern | Brilliant Telecommunications | Randburg

August, 2021 – January 2022

- Supported LAN/WAN for 500+ users: VLAN segmentation, STP/RSTP, DHCP/DNS, and Cisco WLC wireless.
- Configured **FortiGate 80E** for NAT, port forwarding, and site-to-site VPNs; performed quarterly firmware upgrades during maintenance windows.
- Wrote **Python scripts** (pysnmp) to poll interface utilisation and CPU/memory with alerts to Microsoft Teams - eliminated daily manual health checks.
- Assisted in Layer-2 migration from Catalyst 2960 to **Cisco 9300** stack; documented all physical/logical topologies in NetBox.

- Diagnosed complex routing issues: OSPF neighbour flapping (MTU mismatches) and BGP route dampening - implemented corrective route-maps.
- **AI experimentation:** Used a local LLM (Ollama) to summarise syslog bursts during network events - shared with the team as a troubleshooting accelerator.

KEY PROJECTS

MPLS-VPN Lab-to-Production Migration

- Designed a lab environment (EVE-NG) with 6 PE routers, 2 P routers, and 4 CE devices running OSPF + BGP + MPLS LDP + VPNv4.
- Translated validated configs to production using Ansible templates - onboarded 3 new customer VRFs with zero service interruption.

FortiGate Policy Automation Suite

- Developed a Python + fortiosapi tool that scans all policies for shadowed/unused rules, generates a cleanup report with suggested merges, and pushes approved changes via API.
- Reduced rule count by 30% and improved firewall throughput.

AI-Powered BGP Troubleshooter

- Built a Slack bot that accepts BGP outputs, sanitises them, and uses an LLM API to identify flapping prefixes, likely causes (withdraws, next-hop unreachable, AS-path loops), and suggested route-map or prefix-list fixes.
- Adopted by 4 peers in the NOC team.

TECHNICAL SKILLS

Service Provider Routing	BGP (full-table, communities, peer-groups, route-reflectors), OSPF (LSA types, stub/NSSA), MPLS (LDP, RSVP-TE), VPNv4, 6PE, IS-IS (basic), BFD, PIM-SM
Switching / Data Centre	VLAN, 802.1Q, STP/RSTP/MSTP, EtherChannel, vPC, VXLAN, EVPN (type-2/3/5 routes), MAC-VRF
Firewalls (Fortinet)	FortiGate CLI + GUI, FortiManager, policies, NAT, SSL-VPN, IPsec, application control, web filtering, HA clustering; Cisco ASA/Firepower (basic)
Load Balancing	HAProxy (ACLs, backends, stick-tables, stats), NGINX (reverse proxy, caching, rate-limiting)
Automation	Python (netmiko, NAPALM, requests, Jinja2), Ansible (core + fortios collection), Bash, Git, GitLab CI/CD, NetBox/Nautobot
Monitoring	Prometheus, Grafana, LibreNMS, Splunk, ELK (basic), SNMPv3, NetFlow/IPFIX (basic)
AI Tools	GitHub Copilot, ChatGPT, Claude - daily use for code reviews, config generation, and after-action documentation
OS / Virtualisation	Linux (RHEL/Ubuntu), Windows Server, VMware ESXi, Docker, EVE-NG / CML for labbing

EDUCATION

Bachelor of Science in Information Technology - 2020

North West University, Vanderbijlpark, South Africa

Relevant modules: Advanced Networking, Network Security, Data Communications, Scripting for Systems Administration

ADDITIONAL INFORMATION

- **On-site / travel:** Full driving licence; willing to travel between sites (up to 50%) and participate in on-call rotation with change-window availability.
- **Languages:** English (fluent), Isindebele, Shona, Venda, Sotho.

- **Continuous learning:** Currently building an EVPN/VXLAN lab with FRR and VyOS; studying for Fortinet NSE 4.

REFERENCES

Available upon request.